

Limitations and Threats

Sybil Attacks

A Sybil attack occurs when one entity creates multiple identities to manipulate a system.

This project does not prevent users from creating multiple blockchain addresses.

As a result:

- A user could create new accounts to reset reputation.
- The system does not link addresses to real-world identities.

Lack of Identity Verification

The system assumes that each blockchain address represents a distinct user.

However:

- There is no identity verification mechanism.
- There is no authentication beyond wallet ownership.
- The contract evaluates behavior per address, not per real-world individual.

Timestamp Manipulation Risk

The system relies on block.timestamp for decay calculations. While timestamps are generally accurate, validators can adjust them slightly within acceptable limits.

This may cause:

- Minor variation in decay timing.
- Slight differences in interval boundaries.

The design uses coarse time intervals to reduce sensitivity to small timestamp variations.

No Behavioral Quality Verification (for now!)

The system increases reputation based on interaction events. It does not verify:

- Whether interactions are meaningful.
- Whether behavior is beneficial or harmful.
- The quality of contributions.

Reputation reflects activity and time-based consistency, not behavioral quality.(for now)

Contract Immutability Risk

Once deployed, smart contract logic cannot be modified. If a flaw exists in:

- Decay calculation,
- Reward logic,
- Boundary enforcement,

It would require deploying a new contract. For this reason, the system is deployed on a test network and treated as a prototype.

Limited Scalability (too advance)

The contract is not optimized for:

- High-frequency transactions,
- Large-scale user bases,
- Gas-efficient mass updates.

No Economic Incentive Mechanism

The system does not include:

- Token rewards,
- Financial penalties,
- Governance voting.

It focuses solely on reputation tracking and explainability.